

A Threshold-Family Subcase of the Isometry Group Problem

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Status

This is a candidate partial result for the final questions in Brech–Ferenczi–Tcaciuc, *Isometries of combinatorial Banach spaces*, arXiv:1907.06815 [1]. It does not solve the general classification problem. It gives a complete classification for a natural class of regular families defined by a nondecreasing threshold function.

Source Question

In Section 5 of [1], after proving that isometries of combinatorial spaces are signed permutations, the authors ask:

Given a regular family $\mathcal{F}$, what are the permutations of the basis which induce an isometry of $X_{\mathcal{F}}$?

For which combinatorial spaces can we explicitly describe the group of its isometries?

The source crop below is from page 11 of the arXiv PDF.

ISOMETRIES OF COMBINATORIAL BANACH SPACES

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5. FINAL REMARKS

Theorem 10 guarantees that all the isometries of a combinatorial space or its dual are determined by a permutation of the elements of the basis and a change of signs. Natural and general questions which remain open are the following:

Question 16. *Given a regular family $\mathcal{F}$, what are the permutations of the basis which induce an isometry of $X_{\mathcal{F}}$?*

Question 17. *For which combinatorial spaces can we explicitly describe the group of its isometries?*

In Section 4 we described the group of the isometries on Schreier spaces, showing that the identity is the only permutation allowed. The following example illustrates an intermediate situation where more permutations are allowed, though not all of them:

Example 18. *Given an increasing sequence $(k_n)_n$ such that $k_n = 0$. let*

Idea of the Proof

For the threshold families below, each integer n carries a purely combinatorial label: the largest cardinality of a member of the family that contains n . Monotonicity of the threshold function makes this largest cardinality exactly $g(n)$. Therefore any permutation preserving the family must preserve every level set of g . Conversely, if a permutation preserves the levels of g , then moving a finite set can only move its minimum to a point with threshold at least as large as the old one, so membership in the family is preserved.

The Partial Result

Let $g : \mathbb{N} \rightarrow \mathbb{N}$ be nondecreasing with $g(n) \geq 1$ for all n . Define

$$\mathcal{F}_g = \{\emptyset\} \cup \{F \in [\mathbb{N}]^{<\omega} : F \neq \emptyset \text{ and } |F| \leq g(\min F)\}.$$

Let

$$L_m = \{n \in \mathbb{N} : g(n) = m\}.$$

Empty level sets are ignored in products below. Let $\mathbb{T}_{\mathbb{K}}$ denote the scalar sign group: $\{-1, 1\}$ over the reals and the unit circle over the complex numbers.

Theorem 1. *The family $\mathcal{F}_g$ is regular. A permutation π of $\mathbb{N}$ induces an isometry of $X_{\mathcal{F}_g}$, equivalently of $X_{\mathcal{F}_g}^*$, if and only if*

$$g(\pi(n)) = g(n) \quad \text{for every } n \in \mathbb{N}.$$

Consequently every surjective linear isometry of $X_{\mathcal{F}_g}$ or of $X_{\mathcal{F}_g}^$ is a signed permutation preserving the level sets L_m , and*

$$\text{Isom}(X_{\mathcal{F}_g}) \cong \mathbb{T}_{\mathbb{K}}^{\mathbb{N}} \rtimes \prod_{m \geq 1} \text{Sym}(L_m),$$

with the same permutation part for $\text{Isom}(X_{\mathcal{F}_g}^)$.*

Proof. First $\mathcal{F}_g$ is hereditary: if $E \subseteq F \in \mathcal{F}_g$ and $E \neq \emptyset$, then $|E| \leq |F| \leq g(\min F)$, while $\min E \geq \min F$ and g is nondecreasing, so $|E| \leq g(\min E)$. The spreading property is similar: if $F = \{n_1 < \dots < n_k\} \in \mathcal{F}_g$ and $m_i \geq n_i$ with $m_1 < \dots < m_k$, then $k = |F| \leq g(n_1) \leq g(m_1)$. The family contains all singletons because $g \geq 1$.

To prove compactness, view $\mathcal{F}_g$ as a subset of $2^{\mathbb{N}}$. Suppose $F_j \in \mathcal{F}_g$ converges pointwise to $A \subseteq \mathbb{N}$. If A were infinite and $a = \min A$, then for all large j the set F_j would contain a and the first $g(a) + 1$ elements of A , and would contain no integer below a . Hence $\min F_j = a$ and $|F_j| \geq g(a) + 1$, contradicting $F_j \in \mathcal{F}_g$. Thus A is finite. If A is nonempty and $a = \min A$, then for all large j , all points of A belong to F_j and no point below a belongs to F_j ; consequently $\min F_j = a$ and $|A| \leq |F_j| \leq g(a)$. Hence $A \in \mathcal{F}_g$. Therefore $\mathcal{F}_g$ is compact.

Now define the family-invariant number

$$h_{\mathcal{F}_g}(n) = \sup\{|F| : F \in \mathcal{F}_g, n \in F\}.$$

Because g is nondecreasing, any $F \in \mathcal{F}_g$ containing n satisfies

$$|F| \leq g(\min F) \leq g(n).$$

On the other hand, the set $\{n, n+1, \dots, n+g(n)-1\}$ belongs to $\mathcal{F}_g$ and contains n , so $h_{\mathcal{F}_g}(n) = g(n)$.

Let π be a permutation of $\mathbb{N}$ with $\pi(\mathcal{F}_g) = \mathcal{F}_g$. Then cardinalities of family members containing a point are preserved, so

$$g(\pi(n)) = h_{\mathcal{F}_g}(\pi(n)) = h_{\mathcal{F}_g}(n) = g(n).$$

Thus every family automorphism preserves the level sets of g .

Conversely, suppose $g(\pi(n)) = g(n)$ for all n . Take a nonempty $F \in \mathcal{F}_g$ and let $r = \min F$. Let $s \in F$ be such that $\pi(s) = \min \pi(F)$. Since $s \geq r$ and g is nondecreasing,

$$g(\min \pi(F)) = g(\pi(s)) = g(s) \geq g(r) \geq |F| = |\pi(F)|.$$

Thus $\pi(F) \in \mathcal{F}_g$. Applying the same argument to π^{-1} gives $\pi(\mathcal{F}_g) = \mathcal{F}_g$.

Brech–Ferenczi–Tcaciuc prove that for regular families, all isometries of combinatorial spaces and their duals are signed permutations, and that a permutation induces an isometry exactly when it sends the family to itself [1, Theorem 10 and the following corollary]. Combining that theorem with the preceding family-automorphism classification gives the asserted description of $\text{Isom}(X_{\mathcal{F}_g})$ and $\text{Isom}(X_{\mathcal{F}_g}^*)$. $\square$

Relation to the Source Example

The final example of [1] treats the case where the threshold value is constant on consecutive intervals and increases by one from one interval to the next. The theorem above allows arbitrary nondecreasing threshold functions: jumps may skip values, plateaus may be finite or infinite, and the constant case recovers the full symmetric permutation group.

Limitations and Novelty Check

This is a partial result only. It does not classify the isometry groups of all combinatorial spaces, nor does it address regular families whose membership is not determined by the pair $(|F|, \min F)$.

Local duplicate checks searched the run indexes for arXiv:1907.06815, combinatorial Banach spaces, isometry groups, permutations, and threshold families. The source paper itself was searched for threshold-family and level-set language. No existing run packet or attempt was found for this subcase. No external literature search was used, so novelty confidence is local rather than global.

Verification Notes

No computational step is used. The proof reduces the isometry classification to the source theorem of [1] and proves the remaining family automorphism statement directly. The two proof points most worth checking are the compactness argument for $\mathcal{F}_g$ and the converse direction showing that a g -level-preserving permutation sends every $F \in \mathcal{F}_g$ back into $\mathcal{F}_g$.

References

- [1] C. Brech, V. Ferenczi, and A. Tcaciuc, *Isometries of combinatorial Banach spaces*, arXiv:1907.06815.